

Chelpuri Sai Madhav

Software Engineer

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SUMMARY

Software Engineer with a Master's degree in Computer Science and 5 years of experience designing and delivering scalable, cloud-native applications in healthcare, automotive, and fintech. Proficient in Java, Spring Boot, React.js, Angular, GraphQL, and AWS, with a track record of proven results by reducing claim processing latency by 60%, improving system uptime to 99.9%, and accelerating deployments from bi-weekly to daily. Skilled in microservices, event-driven architectures, and CI/CD automation with Docker, Jenkins, and Kubernetes, delivering secure, scalable, and compliant solutions. Strong collaborator with experience mentoring teams, working in Agile teams, and driving features that improve both user experience and business outcomes.

SKILLS

Methodologies: Agile, Scrum, Code Reviews, Technical Documentation

Programming Languages: C/C++, Java/J2EE, JavaScript, TypeScript, Python, SQL

Java Technologies: JDBC, Servlets, JSP, JDK, MOCKITO, RESTEasy

Backend Frameworks: Spring Boot, Spring MVC, Spring Security, Hibernate, Flask, Microservices, REST APIs, Kafka

Frontend Frameworks: React.js, Redux, Angular, Node.js

Containers and Orchestration Tools: Docker, ECS, EKS

IDE's: IntelliJ IDEA, VS Code, Chrome DevTools

Application Server: Apache Tomcat, Apache Kafka

Web Technologies: HTML, CSS3, Bootstrap, Tailwind CSS, JSON, XML

Databases: MySQL, PostgreSQL, MongoDB, Database Design, Query Optimization

CI-CD/Build Tools: Jenkins, Maven, Git, CI/CD Pipelines, JUnit, Jest, Webpack

Version Control Tools: Git, GitHub, Jira, BitBucket

Cloud: AWS (EC2, S3, Lambda, RDS, CloudWatch, IAM), Azure

Security: OAuth2, JWT, API Security, HTTPS

Testing: JUnit, Jest, Postman, Integration Testing

EXPERIENCE

Software Engineer

CVS Health | Birmingham, AL, US

Aug 2023 – Present

- Developed React.js and Redux-based pharmacy benefits modules for claims workflows, reducing pharmacist task time by 30% and supporting 50K+ daily users across 200+ retail locations.
- Built and deployed Spring Boot microservices for eligibility and prior authorization, cutting mainframe dependencies by 40% and increasing throughput from 5K to 12K requests per hour.
- Implemented OAuth2, JWT, and Spring Security for HIPAA-compliant role-based access, protecting 100K+ patient records and ensuring PHI data governance.
- Led GraphQL migration for patient dashboards, eliminating 60% of redundant REST calls and reducing page load latency by 2s, boosting member satisfaction and lowering call center volume by 15%.
- Engineered secure Rx billing workflows with ISO 8583 protocols and fraud detection, processing \$5M+ in monthly transactions while maintaining PCI-DSS compliance and reducing billing errors by 25%.
- Automated CI/CD pipelines with Jenkins, Docker, and AWS CodePipeline, raising deployment frequency from bi-weekly to daily and reducing production incidents by 35% while sustaining 99.9% uptime.
- Established monitoring and alerting with AWS CloudWatch and Datadog, creating proactive dashboards that reduced MTTR by 40% and prevented 20+ potential outages per month.
- Collaborated with QA, compliance, and product teams in Agile cycles, delivering 95% of planned features on schedule and raising test coverage from 65% to 85% through peer code reviews.
- Refactored legacy prescription management modules and implemented automated testing, cutting technical debt by 30% and preventing 25+ production defects.

Environment: React.js, Redux, TypeScript, Java 17, Spring Boot, Spring Security, GraphQL, OAuth2, JWT, PostgreSQL, Redis, AWS (EC2, S3, CloudWatch, CodePipeline), Jenkins, Docker, ISO 8583, Datadog, JUnit, Jest, Agile/Scrum

Software Engineer

Mercedes-Benz Research and Development | Bengaluru, India

Jan 2022 - Jan 2023

- Engineered Angular-based telemetry dashboards for MBUX Digital Cockpit systems, integrating Redis caching and optimized data binding to cut HMI latency by 15% and deliver smoother, real-time driver interface experiences in test vehicles.
- Implemented 3 Spring Boot microservices on AWS ECS, processing 100K+ daily diagnostic requests and reducing system response latency from 5s to 1.2s for vehicle health monitoring systems.
- Assisted in configuring Kafka consumer services to handle 500K+ daily telematics events, building transformation pipelines that achieved sub-2s latency for real-time dashboard analytics.
- Refactored Spring Boot safety alert components with asynchronous processing, reducing alert delivery failures by 20% and improving reliability in safety-critical scenarios.

- Integrated OAuth2 and JWT authentication across connectivity APIs, enforcing GDPR-compliant access controls and reducing unauthorized access incidents by 35%.
- Containerized microservices using Docker and collaborated with DevOps teams on Kubernetes manifests, cutting environment provisioning time from 4h to 15m and ensuring consistent test deployments.
- Partnered with Hardware-in-Loop testing teams to resolve dashboard integration issues, enhancing Splunk logging and creating diagnostic documentation that reduced incident resolution time by 30%.

Environment: Java, React.js, Angular, TypeScript, Spring Boot, Kafka, Redis, PostgreSQL, Docker, Kubernetes, AWS (ECS, Lambda, S3), OAuth2, JWT, Jenkins, GitLab CI/CD, Splunk

Java Application Developer

Accenture | Hyderabad, India

Jan 2021 - Dec 2021

- Contributed development of customer onboarding and KYC compliance modules using React.js and JavaScript for a Tier-1 banking client, delivering mobile-responsive interfaces that reduced onboarding abandonment by 15%.
- Developed RESTful web services integrating backend systems with Experian and CIBIL credit bureau APIs, supporting real-time loan validations and reducing manual underwriting turnaround from 24h to 2h for 1,000+ daily applications.
- Optimized UI performance by refactoring CSS layouts and implementing jQuery lazy loading, cutting rendering times by 25% across loan application portals and improving customer experience scores.
- Automated deployments to Azure App Services with Jenkins CI/CD pipelines, eliminating 15+ manual release steps and achieving a 99.5% deployment success rate in dev and staging environments.
- Implemented unit testing with TestNG and supported automated builds, maintaining 85% coverage and preventing 20+ regression defects during sprint releases.
- Collaborated with global teams using Bitbucket and JIRA, contributing to 30% faster defect resolution and improving sprint velocity by 10% through code reviews and knowledge sharing.
- Enhanced backend resilience by adding exception handling and retry logic for third-party API integrations, reducing transaction failures by 18% during credit bureau outages.
- Worked under mentorship of senior developers on Java best practices, secure coding, and PCI-DSS, KYC compliance, contributing to client demos and production deployments.

Environment: Java, Spring Framework, AngularJS, JavaScript, HTML5, CSS3, jQuery, REST APIs, Azure App Services, Jenkins, TestNG, Bitbucket, JIRA, Experian APIs, CIBIL integration, PCI-DSS compliance

PROJECTS

AI-Enabled Helpdesk Automation Chatbot

- Built a Java-based chatbot with Spring Boot, OpenAI API, and AWS Lambda, enabling context-aware responses that reduced support ticket escalations by 40% and improved customer resolution times.
- Deployed scalable microservices on AWS (Lambda, EC2, DynamoDB, S3) with NLP-powered query handling, delivering low-latency interactions and ensuring 99.9% availability for enterprise users.

Customer Risk Assessment Module

- Built React dashboards and Spring Boot scoring services to evaluate patient risk, delivering sub-500ms API responses for real-time compliance assessments and improving decision-making speed.
- Integrated GraphQL for unified patient data and automated deployments with Jenkins/Docker, cutting dashboard load times and reducing iteration cycles by 40%.

EDUCATION

- **University of Alabama at Birmingham, US**
Master of Science in Computer Science.
- **Vellore Institute of Technology, India**
Bachelor of Technology, Computer Science Engineering

CERTIFICATIONS

- AWS Certified Solutions Architect - Associate.
- Oracle Certified Professional, Java SE 8 Developer.
- Microsoft Certified: Azure Developer Associate.